

Reviewer Pack — Minimal Rank of Quasi-vertex-Descent Trees over Tseitin Circ...

Entry 35648c0d5a85 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quasi-vertex-Descent Trees over Tseitin Circuit Size

Entry ID: 35648c0d5a85 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 14:26:09 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Quasi-vertices) **Field B** (complexity object): Complexity Theory: Tseitin Circuit Size

Statement:

[‘For a given Tseitin circuit, the minimal rank of its associated quasi-vertex-descent tree is at least $2^{\Omega(\log n)}$.’, ‘The associated quasi-vertex-descent tree is constructed by considering each gate in the Tseitin circuit and associating it with a vertex. Edges between vertices are drawn based on the input-output relationships of the gates, forming a tree structure.’, ‘Quasi-vertices refer to non-standard points that include both vertices and edges from the original graph.’]

Rationale (proposer’s reasoning):

[‘Algebraic combinatorics provides a rich set of tools to study the complexity of computational problems through the lens of algebraic objects. Quasi-vertex-descent trees provide a structured way to represent Tseitin circuits, potentially revealing underlying complexities.’, ‘The conjecture suggests that there is a deep connection between the structure of these quasi-vertex-descent trees and the size of the Tseitin circuit, which could lead to new insights into resolution complexity.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c46d283c986580f1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the conjecture that the minimal rank of quasi-vertex-descent trees over Tseitin circuit size is at least $2^{\Omega(\log n)}$, a result will be considered supported if, for at least 80% of the 30 randomly chosen instances with $n \leq 40$, the calculated minimal rank of the associated quasi-vertex-descent tree is $\geq 2^{(0.5 \log n)}$. A result will be falsified if any instance has a minimal rank $< 2^{(0.5 \log n)}$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_tseitin_circuit(n):
        if n == 1:
            return "x"
        else:
            a = generate_tseitin_circuit(n // 2)
            b = generate_tseitin_circuit(n - n // 2)
            return f"({a} | {b})"

    def construct_quasi_vertex_descent_tree(circuit):
        if isinstance(circuit, str):
            return circuit
        else:
            left = construct_quasi_vertex_descent_tree(circuit[0])
            right = construct_quasi_vertex_descent_tree(circuit[2])
            return f"({left} {circuit[1]} {right})"

    def calculate_minimal_rank(tree):
        if isinstance(tree, str):
            return 1
        else:
            left_rank = calculate_minimal_rank(tree[0])
            right_rank = calculate_minimal_rank(tree[2])
            return max(left_rank, right_rank) + 1

    n = random.randint(5, 40)
    circuit = generate_tseitin_circuit(n)
    tree = construct_quasi_vertex_descent_tree(circuit)
```


The test has only been conducted on a very small number of instances ($n \leq 15$). The conjecture suggests that the minimal rank should scale with n , but the evidence provided does not show this scaling. Additionally, there is no indication that the metric definition bug or construction gap issues have been addressed.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that for at least one instance (rank=1), the minimal rank of the associated quasi-vertex-descent tree is less than the expected | next: Further investigation is needed to understand why the minimal rank is consistently lower than expected and to refine the conjecture or its supporting evidence.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11893
2	preregistration	ollama_remote	glm4:latest	0	0	6419
3	novelty	ollama_remote	glm4:latest	0	0	4942
4	novelty	ollama_remote	glm4:latest	0	0	5098
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	39578
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8234
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7244
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7613
9	critic	ollama_remote	glm4:latest	0	0	11762
10	judge	ollama_remote	glm4:latest	0	0	5721

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 108504 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/35648c0d5a85.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/35648c0d5a85.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/35648c0d5a85.tar.gz` (if generated)